

Timothy Carroll

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EDUCATION

University of California San Diego

Sep. 2025 - June 2028

Bachelor of Science in Electrical Engineering

College of Marin

Aug. 2023 – May 2025

- 3.8 GPA, Honor Roll 2023 - 2025
- Member of the Engineering Club, Vice President of the Physics Club

EXPERIENCE

Broadcast Engineering Intern

June 2025 - August 2025

KNTV NBC Bay Area

San Jose, CA

- Supported the Technical Operations Center (TOC) by routing live video feeds from field crews, helicopters, and studio cameras using Dalet Galaxy workflows.
- Assisted with live production workflows including clipping and recording sports games, press conferences, and breaking news events.
- Shadowed engineering staff to troubleshoot issues and helped build workflow redundancies that reduced points of failure during broadcasts.

MESA Program Tutor

Aug. 2024 - May 2025

College of Marin

Kentfield, CA

- Founded and led the program's weekly Physics Study Group, creating a structured and supportive learning environment; Recognized with Employee of the Month and Service to Community awards.
- Sole physics tutor providing individual and group instruction across the full physics sequence: Mechanics/Properties of Matter, Electricity & Magnetism, Modern Physics(Heat, Light & Sound)

Nanofabrication Research Intern

May 2024 - Aug. 2024

Carnegie Mellon University

Pittsburgh, PA

- Researched lithographic and oxide etch processes for semiconductor fabrication, optimizing resolution and process fidelity.
- Analyzed resist thickness vs. spin speed across multiple resist types and developed high-selectivity oxide etch and lift-off processes.
- Operated e-beam lithography, sputtering systems, profilometers, and nanospectrometers in a cleanroom environment.
- Designed a centralized spin-table data system adopted lab-wide and presented research findings to mentors and industry professionals.

Audio Design Intern

Dec. 2023 - Jan. 2024

Roswell Pro Audio

Santa Rosa, CA

- Beta tested a custom high-voltage power supply for audio hardware and diagnosed AC noise through signal-path analysis.
- Measured low-level AC noise using oscilloscopes and spectrum analyzers while maintaining high-voltage safety.
- Evaluated hardware layout and signal integrity through teardown, analysis, and troubleshooting.

PROJECTS

8-Bit Computer Built from Logic Gates

July 2025 - Present

- Building an 8-bit computer from discrete logic gates, including registers, an RC-based clock, program counter, and ALU.
- Constructed two registers and a program counter on hand-wired breadboards; designed the clock circuit to control pulse width.
- Debugged power and noise issues using power-rail capacitors, pull-up resistors, current-limiting modifications, and soldered circuit layouts.

Sallen-Key Active Low-Pass Filter

May 2025

- Designed a second-order Sallen-Key active low-pass filter to separate DSL data from voice signals on a telephone line.
- Selected component values for a controlled passband response and simulated non-ideal op-amp effects using LTspice.
- Built and tested the filter experimentally, comparing measured frequency response with analytical and MATLAB results.

12-Bit Sequence Recognizer FPGA – SONY Remote

Nov. 2024

- Designed a sequential circuit using D flip-flops and shift registers to recognize SONY TV remote control bit sequences.
- Developed Verilog RTL for 4-bit and 12-bit shift registers and implemented the design on a Basys FPGA board.
- Interfaced the FPGA with a breadboarded infrared receiver to capture and process remote-control signals.

TECHNICAL SKILLS

Hardware: Oscilloscope, Signal Generator, Multimeter, Soldering & Prototyping, DC/AC Circuit Analysis, Filter Design, Combinational & Sequential Logic

Software: C/C++, Embedded Systems Programming, Verilog (RTL), LTspice, KiCad (PCB Design), Fourier & Laplace Analysis, Convolution & LTI Systems